Experimental techniques and chemical analysis

Igcse Chemistry

Experimental design

Apparatus

You should know which piece of apparatus to use for each measurement:

Measurement	Apparatus
time	**stop-watch** 秒表
temperature	**thermometer** 温度计
mass	**balance** 天平
volume (accurate)	**burette** 滴定管 or **volumetric pipette** 移液管
volume (rough)	**measuring cylinder** 量筒
volume of a gas	**gas syringe** 注射器

Key words

These words are used throughout practical chemistry:

- A **solvent** 溶剂 is a substance that dissolves a solute.
- A **solute** 溶质 is the substance that dissolves in the solvent.
- A **solution** 溶液 is a mixture of one or more solutes dissolved in a solvent.
- A **saturated solution** 饱和溶液 holds the maximum amount of solute that will dissolve at a given temperature.
- A **residue** 残渣 is the substance left behind after evaporation, distillation or filtration.
- A **filtrate** 滤液 is the liquid that has passed through a filter.

Acid–base titrations

A **titration** 滴定 finds the exact volume of one solution that reacts with another.

- Use a **volumetric pipette** to measure a fixed volume of one solution into a flask.
- Add a few drops of a suitable **indicator** 指示剂.
- Add the other solution from a **burette**, slowly, until the colour just changes.

The **end-point** 终点 is the moment the indicator changes colour, which shows the reaction is exactly complete. You read the burette to find the volume added.

Chromatography

Paper chromatography 纸色谱法 separates a mixture of soluble substances. You put a spot of the mixture near the bottom of the paper, then stand the paper in a solvent. As the solvent rises up the paper, the substances move different distances, so they separate.

- For **coloured** substances you can see the spots directly.
- For **colourless** substances you must spray a **locating agent** 显色剂 to make the spots show up.

The finished paper is a **chromatogram** 色谱图. You can use it to:

- identify an unknown substance by comparing it with known substances;
- tell if a substance is pure (a pure substance gives only one spot).

You can also calculate the R_{f} value of a spot:

$$R_{\mathrm{f}} = \frac{\text{distance moved by the substance}}{\text{distance moved by the solvent}}$$

Separation and purification

You choose a method based on the mixture:

Method	Used to separate
dissolving in a suitable solvent, then **filtration** 过滤	an insoluble solid from a liquid
crystallisation 结晶	a soluble solid from its solution
simple distillation 蒸馏	a solvent (the liquid) from a solution
fractional distillation 分馏	two or more liquids with different boiling points

You can check the purity of a substance using its **melting point** 熔点 and **boiling point** 沸点: a pure substance melts and boils at sharp, fixed temperatures, while impurities lower the melting point and raise the boiling point.

Identifying ions and gases

Tests for anions

An **anion** 阴离子 is a negative ion.

Anion	Test	Result
carbonate 碳酸盐 (CO_3^{2-})	add dilute acid	fizzes; the gas turns limewater milky (**carbon dioxide** 二氧化碳)
chloride 氯化物 (Cl^-)	add dilute **nitric acid** 硝酸, then **silver nitrate** 硝酸银	white precipitate
bromide 溴化物 (Br^-)	add dilute nitric acid, then silver nitrate	cream precipitate
iodide 碘化物 (I^-)	add dilute nitric acid, then silver nitrate	yellow precipitate
nitrate 硝酸盐 (NO_3^-)	add **aluminium** 铝 foil and **sodium hydroxide** 氢氧化钠, warm	**ammonia** 氨气 gas given off
sulfate 硫酸盐 (SO_4^{2-})	add dilute nitric acid, then **barium nitrate** 硝酸钡	white precipitate
sulfite 亚硫酸盐 (SO_3^{2-})	add acidified **potassium manganate(VII)** 高锰酸钾	purple colour fades

Tests for cations

A **cation** 阳离子 is a positive ion. Add aqueous sodium hydroxide, or aqueous ammonia, and look at the **precipitate** 沉淀 formed.

Cation	With sodium hydroxide	With aqueous ammonia
aluminium (Al^{3+})	white, dissolves in excess	white, stays
ammonium 铵 (NH_4^+)	ammonia gas when warmed	—
calcium 钙 (Ca^{2+})	white, stays	no precipitate
chromium(III) 铬 (Cr^{3+})	green, dissolves in excess	green, stays
copper(II) 铜 (Cu^{2+})	light blue, stays	light blue, dissolves to deep blue
iron(II) 铁 (Fe^{2+})	green, stays	green, stays
iron(III) (Fe^{3+})	red-brown, stays	red-brown, stays
zinc 锌 (Zn^{2+})	white, dissolves in excess	white, dissolves in excess

Tests for gases

Gas	Test	Result
ammonia (NH_3)	damp red **litmus** 石蕊 paper	turns blue
carbon dioxide (CO_2)	bubble through **limewater** 石灰水	turns milky
chlorine 氯气 (Cl_2)	damp litmus paper	bleached white
hydrogen 氢气 (H_2)	a lighted splint	burns with a squeaky pop
oxygen 氧气 (O_2)	a glowing splint	relights
sulfur dioxide (SO_2)	acidified potassium manganate(VII)	purple colour fades

Flame tests

A **flame test** 焰色试验 identifies some metal cations by the colour they give to a flame:

Cation	Flame colour
lithium 锂 (Li^+)	red
sodium 钠 (Na^+)	yellow
potassium 钾 (K^+)	lilac (purple)
calcium (Ca^{2+})	orange-red
barium 钡 (Ba^{2+})	light green
copper(II) (Cu^{2+})	blue-green